

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I D. Dhaubana (192524243) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D. Dhaubana

level 3

10

Dharshana D | Register Number : 192524243

1. Formulate the 8-puzzle problem (a 3×3 sliding tile puzzle) as a search problem by specifying the state representation. Explain why the branching factor of the problem is at most 4.
2. A hospital wants to build an AI agent to triage incoming patients based on symptom severity. Define an appropriate performance measure for this agent and explain the concept of rationality in this context.
3. A warehouse company wants to deploy robots to pick and transport goods to packing stations. Identify which agent architecture is most appropriate + justify your choice with reasons.

ASSESSMENT TOOL - 1

8-Puzzle as a Search Puzzle problem

- A state is represented by a 3×3 grid containing eight numbered tiles (1-8) and one blank space (-).

Eg:

1	2	3
4	5	6
7	-	8

Initial state:

- The starting arrangement of the tiles provided to the problem.

Eg:

2	8	3
1	6	4
7	-	5

Goal Test:

1	2	3
4	5	6
7	8	-

Possible Actions:

- Move the blank space
 - Up
 - Down
 - Left
 - Right

Branching Factor:

- The branching factor is the maximum number of successor states from any state.
- In a 3×3 puzzle, the blank tile can have:
 - 2 moves when at a corner
 - 3 moves when on an edge
 - 4 moves when in the centre.
- $\therefore$, the maximum branching factor is 4.

2] AI Agent for Hospital Patient Triage

Performance measure: The performance measure should evaluate how effectively the agent prioritizes patients.
Possible criteria include:

- Accuracy of severity classification.
- Minimizing waiting time for critical patients
- Reducing risk of delayed treatment
- Efficient use of medical resources
- Patient safety and treatment outcomes.

Rationality: A rational agent selects the action that maximizes the expected value of its performance measure based on available information

In this context, the agent's action is rational if:

- It correctly analyzes the patient's symptoms, vital signs and medical history.
- It assigns the patient to the appropriate priority level.
- It chooses the action that is most likely to improve outcomes and reduce risk.

Eg: If a patient reports chest pain, breathing difficulty, and abnormal vital signs, assigning high priority / emergency care is a rational action because it maximizes patient safety.

3. AI Agent for Warehouse Robots

Most Appropriate Architecture: Learning Agent

Reasons:

1. Dynamic Environment
 - Warehouse layouts, inventory locations, and traffic conditions may change.
 - A learning agent can adapt to these changes over time.
2. Performance Improvement
 - The robot can learn shorter routes and better picking strategies from experience.

3. Obstacle Handling

- learns to avoid congestion and unexpected obstacles efficiently.

4. Optimization.

- Improves task completion time and reduces energy consumption.

5. Scalability.

- Can adapt when new products, shelves, or packing stations are added.

Why not other architectures?

- Simple reflex agent : Reacts only to current perception cannot adapt.
- Model-based Agent: Maintains environment information but does not learn.
- Goal-based Agent: Reaches goals but may not optimize performance.
- Utility-based Agent: Chooses best action using utility values but still lacks learning capability
- Learning agent: Combines goal achievement, optimization + adaptation, making it most suitable for warehouse operations.